

Auteur: Kyan Van den Eynde

Studierichting: Informatica

Oefeningenreeks 3B nr. 3

$$f(x) = \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

$$f'(x) = -\frac{1}{2}x^{-\frac{3}{2}}$$

$$f''(x) = \left(-\frac{1}{2} \cdot \left(-\frac{3}{2}\right)\right)x^{-\frac{5}{2}} = \frac{3}{4}x^{-\frac{5}{2}}$$

$$f'''(x) = \left(\frac{3}{4} \cdot \left(-\frac{5}{2}\right)\right)x^{-\frac{7}{2}} = -\frac{15}{8}x^{-\frac{7}{2}}$$

$$f^{iv}(x) = \left(-\frac{15}{8} \cdot \left(-\frac{7}{2}\right)\right)x^{-\frac{9}{2}} = \frac{105}{16}x^{-\frac{9}{2}}$$

References